

6. (1) 数字组成的非空字符串

(2) 0127

最左: $N \rightarrow ND \rightarrow NDD \rightarrow NDD \rightarrow DDD \rightarrow DDD \rightarrow 01DD \rightarrow 012D \rightarrow 0127$
 最右: $N \rightarrow ND \rightarrow N7 \rightarrow ND7 \rightarrow N27 \rightarrow ND27 \rightarrow \overline{ND27} \xrightarrow{N127} \overline{ND27} \xrightarrow{D127} 0127$

568

最左: $N \rightarrow ND \rightarrow NDD \rightarrow DDD \rightarrow 5DD \rightarrow 56D \rightarrow 568$

最右: $N \rightarrow ND \rightarrow N8 \rightarrow ND8 \rightarrow N68 \rightarrow D68 \rightarrow 568$

43

最左: $N \rightarrow ND \rightarrow DD \rightarrow 4D \rightarrow 43$

最右: $N \rightarrow ND \rightarrow N3 \rightarrow D3 \rightarrow 43$

8. (1) $i+i*i$

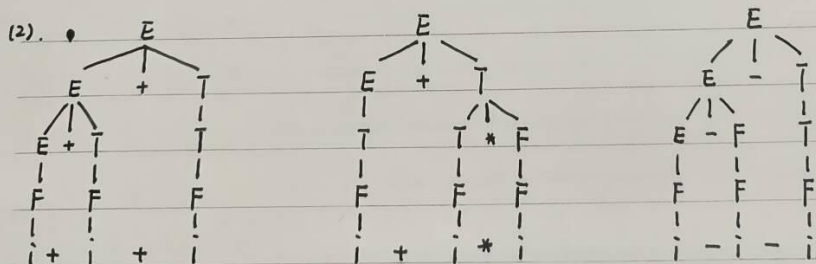
最左: $E \rightarrow E+T \rightarrow T+T \rightarrow F+T \rightarrow i+T \rightarrow i+T^*F \rightarrow i+T^*F \rightarrow i+i^*F \rightarrow i+i*i$

最右: $E \rightarrow E+T \rightarrow E+T^*F \rightarrow E+T^*i \rightarrow E+F^*i \rightarrow E+i^*i \rightarrow T+i^*i \rightarrow F+i^*i \rightarrow i+i^*i$

$i*(i+i)$

最左: $E \rightarrow T \rightarrow T^*F \rightarrow F^*F \rightarrow i^*F \rightarrow i^*(E) \rightarrow i^*(E+T) \xrightarrow{T} i^*(E+T) \rightarrow i^*(F+T) \rightarrow i^*(i+T) \rightarrow i^*(i+F) \rightarrow i^*(i+i)$

最右: $T \rightarrow T^*F \rightarrow T^*(E) \rightarrow T^*(E+T) \rightarrow T^*(E+F) \rightarrow T^*(E+i) \rightarrow T^*(T+i) \rightarrow T^*(F+i) \rightarrow T^*(i+i) \rightarrow F^*(i+i) \rightarrow i^*(i+i)$



9. 测试句子: iei

~~推导1: $S \rightarrow iS_1eS_2, S_1 \rightarrow i, S_2 \rightarrow e$~~

~~推导2: $S \rightarrow iS_1, S_1 \rightarrow iS_2eS_3$~~

外层匹配: e 与第1个 i 结合, $(i)(i)(e)(i)$

内层匹配: e 与第2个 i 结合, $(i)(ie)(i)$

10. $S \rightarrow TS|T, T \rightarrow (S)|()$